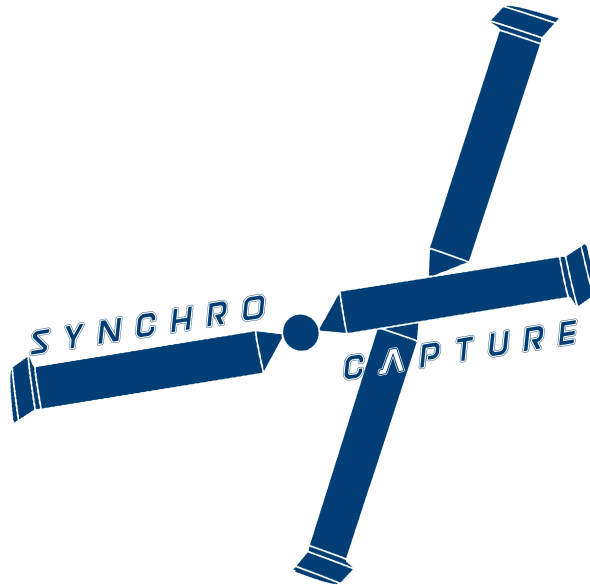

Wholly Reusable Launch System (Recovery)

Project Coordinator



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Abstract

This report summarises my work as one of the project coordinators of the wholly reusable launch system recovery team. My tasks throughout the 5 weeks required team management and leadership, as well as providing a competent level of communication with the launch project as our design depends on them. I also worked on providing the team with the required regulations for our design and created the ground control station for our vehicle. Engineering considerations and common sense were the driving factor when managing the team. We believed that minimal interventions in design decisions was optimal. A stationary ground control station to be operated from land was determined to be the best option for our mission.

Acknowledgements

I would like to thank all the project supervisors for their dedicated work throughout. Their input helped us create as best a design as we could have in the given time frame. Above all, i would like to thank Dr Buxton for his constant help to the coordination team, His advice and help during the 5 weeks of the project greatly facilitated our role as managers and ensured the completion of the project.

I would also like to thank all the individual members of our team. Their relentless and enthusiastic work throughout the 5 weeks contributed to a great design that we are all very proud of. For this project to be a success, all individuals were required to have a significant input, and this was indeed the case, so thank-you to all for this.

WRR Team Members

Table 1: *Team Members and Roles. Sub-team leads are shown in **bold**.*

Name	Team Name	Team Code
Alessandro Bullitta	Project Co-ordination	WRR01
Yassine Ghenima	Project Co-ordination	WRR01
Cyril Gliner	Project Co-ordination	WRR01
Alessandro Rossi Polvara	Aerodynamics	WRR02
Tin Hang Un	Aerodynamics	WRR02
Le Yin	Aerodynamics	WRR02
Chi Wing Cheng	Aerodynamics	WRR02
Alastair Wood	Structures	WRR03
Anton Le	Structures	WRR03
Tejasva Malhotra	Structures	WRR03
Atri Sharma	Structures	WRR03
Omar Arangath	Systems & Propulsion	WRR04
Sam Jenner	Systems & Propulsion	WRR04
Thomas Cross	Systems & Propulsion	WRR04
Jo Tan	Systems & Propulsion	WRR04
Zuzanna Janusz	Systems & Propulsion	WRR04
Tom Walker	Systems & Propulsion	WRR04
Andrew Ng	Flight Mechanics & Controls	WRR05
Kai Hon	Flight Mechanics & Controls	WRR05
India Tavares	Flight Mechanics & Controls	WRR05
Filip Bastien	Flight Mechanics & Controls	WRR05
Ananya Dubey	Flight Mechanics & Controls	WRR05
Tom Alston	Simulations	WRR06
Hakan Serpen	Simulations	WRR06
Tze Leung Wong	Simulations	WRR06

Final Numerical Values

Table 2: *Summary of Key the Specifications for SynchroCapture*

Parameter	Value (Metric)	Value (Imperial)
Global Mass & Power		
Weight with Payload (MTOW)	2907 kg	6414 lb
Empty Weight, Helicopter	1229 kg	2709 lb
Fuel Capacity	278 kg	619 lb
Payload Mass	1400kg	3086 lb
Power Required (Ceiling, ISA)	1193.12 kW	1600 hp
Rotor (For one Rotor)		
Number of Blades	2	2
Blade Mass (each per blade)	40 kg	88 lb
Rotor Diameter	8.05 m	26.4 ft
Rotor Aspect Ratio (AR)	14	14
Rotor Solidity	0.0909	0.0909
Rotor Disk Loading	45.9 kg/m ²	9.4 lb/ft ²
Rotor Blade Loading	503 kg/m ²	103 lb/ft ²
Rotor Hover Tip Speed	205.4 m/s	673.9 ft/s (≈ 0.65 Mach at 20000ft)
Rotor Cruise Tip Speed	268.529 m/s	881 ft/s
Rotational Speeds		
Engine Output Shaft	1256.6 rad/s	12000 RPM
Rotor Speed	51.1 rad/s	487.5 RPM
Travel Speeds		
Cruise Speed	185 km/h, 51 m/s	99 kts
Cruise Speed (Loaded)	100 km/h, 27 m/s	54 kts
Max Speed (Tip Mach = 0.85)	225 km/h, 62 m/s	121 kts
Never Exceed Speed, v_{ne} (Tip Mach = 0.92)	243 km/h, 67 m/s	131 kts
Speed (Supersonic Flow, Tip Mach = 1)	264 km/h, 73 m/s	143 kts
Capture Velocity	144 km/h, 40 m/s	77 kts
Timings		
Total Mission Time	1 hr 49 mins	1 hr 49 mins

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1 Introduction

In this report, I will summarise the roles I undertook throughout this 5-week project of the design of a recovery vehicle for a wholly reusable launch system. My primary role was to coordinate this project alongside 2 other classmates. We would ensure weekly tasks were clear to everyone as well as preparing the layout and time constraints for each weekly meeting. All communication of key values required from the launch team would be communicated to us from their project managers which allows us to pass on this information to our team. I was in charge of all regulations required for our vehicle, with the responsibility of enforcing the key constraints required for our design. Lastly, my side project required me to create a ground control station to operate our UAV from.

2 Project Coordination

To begin the project, myself and the other 2 coordinators got together and decided what the best way to run this project would be. We decided that all 3 of us would have an equal input into the coordination of the project as opposed to having just 1 of us focus on this. We believed this to be optimal as it allows for a variety of opinions when making key engineering decisions about the design and having a second and third opinion to things allows for a more fluid flow of the project and less bias. It was then decided that Cyril would undertake the role of creating a business case for the project, alessandro would work on the cost and I would be doing regulations. Each of us would then go on to do a specific side project relating to the mission.

One of the challenges we were faced with as project coordinators was the fact we had to do this project remotely due to the impact of the coronavirus. The biggest challenge here was the different time zones people were in. Having people in the team that are in a timezone of up to BST+8 proved a challenging. This was a particular issue when it came to meetings. Had something come up quite late during the day, meetings with the people working on this topic would be organised. This would be particularly troublesome if one of the leads in the specific topic was at a BST+8 timezone, as this person would most likely be sleeping. This would mean the problem cannot be solved until this person woke up, but even then, at BST+8 this person could wake up when people in the local timezone would go to sleep. This sets us back multiple hours on the project. This scenario happened on a few occasions. Remote working also made it much harder to communicate than if it was done on campus. Not being able to all be physically present to discuss the design proved hard for such a large project. This was down to some people having poor internet connection, people being unavailable at certain times of the day and simply just not having the physical presence of the team members.

2.1 Initial Organisation Of The Project

We decided to organise the work such that the communication between each sub team would be optimal. Microsoft teams will be our primary means of communication. All relevant meetings and presentations would occur on this platform. All relevant design parameters and configurations would be uploaded on teams.

First, we created a channel for each sub team. These channels would include each member of the sub team and the 3 project coordinators. Secondly, we created a general channel which includes all individuals working on the project and all the project supervisors. The primary use of this channel is for presentation preparation, important announcements regarding major changes or issues with

the design, team name selection etc. Furthermore, we asked each sub team to select a sub team coordinator. The job of this individual is to report to us coordinators about any issues that arise in their part, or about any information they needed from other sub teams or the launch team. We created a channel including all sub team coordinators and project coordinators that would be used as a summary platform regarding the work completed by each sub team. At the end of every day, each sub coordinator would update us, in a couple of brief sentences, about the work completed by all individuals of that sub team. This allowed us coordinators to have a clear image of what was happening throughout the project.

As the project progressed, we had to adapt to the fact that some designs would require the input of multiple sub teams, so whenever this was the case, we simply created another sub channel for that specific design and added all individuals working on this case to this channel. An example of this is, at the beginning of week 3, preliminary work on the fuselage design began, so a channel including all working on this design was created.

2.2 Weekly Progress

At the beginning of each week, myself and the coordinators would get together and prepare a document with all the tasks we would like to be completed by the end of the week by each sub team and in time for the weekly presentation. We would brainstorm ideas as to what we thought would be appropriate topics to be worked on. We followed the general criteria of an aerospace vehicle design, so beginning by working on the baseline configuration, followed by the conceptual design and the preliminary design. Following the baseline configuration, the general way to go about this would be to begin with the conceptual design, and when this is finished, you would proceed to do the preliminary design. In our case however, as our vehicle is a rather unconventional design, we did not do it in this order. We began by doing conceptual design, however, between the second and third week, these 2 design phases started to overlap, meaning we were doing parts of them at the same time. We made sure tasks were doable within the week and did not rely on future parameters (kept tasks in an appropriate order).

2.2.1 Vehicle Selection

The whole of the first week was devoted to the selection of what recovery vehicle we were going to use. This week would prove to be different to the rest of the project. This is because the whole team was split into 3 teams, independent of their individual sub teams. The reason for this is because we had to decide whether we were going to use an aircraft, helicopter, or VTOL (vertical take-off and landing) as a recovery vehicle. 1 coordinator was designated to be the lead coordinator of each of these 3 concepts, I was leading the aircraft concept.

During the first meeting, I told everyone in the team to just familiarise themselves with how an aircraft would go about recovering a large payload. I asked everyone to do some research on various methods of how to recover the payload. I set up a daily morning meeting with the whole team to discuss their progress and to have them discuss their finding and ideas with other group members. This allowed the team to brainstorm a few ideas together and come up with 3 designs we thought would be most suited to our mission.

Towards the tail end of the week, having come up with our top 3 designs, I split the team up and had people working on the Pros and Cons of each configuration, initial sizing of the vehicle, the

cost of operating such vehicle, key data such as the range and manoeuvrability, and finally, I had someone work on a cad model of the designs. These factors would prove paramount during the decision process. The team also spent some time looking into some exotic ideas, however these were very quickly overlooked due to many complications such as the feasibility of the design and the likelihood of success of the mission.

For the decision process, each team chose their best design, and me and the coordinators created a decision matrix outlining all the factors we were going to compare. Each factor was compared to all other factors and was given a score of 1 to 10 regarding its relative importance to what it was being compared to. We then took a normalized mean, and this showed us the most to least important factors when taking the decision. A risk analysis was then performed to determine which of the 3 vehicles was least risky. This gave us a tied score between the VTOL and helicopter. To decide, we compared the two most critical factors as per our decision matrix, the likelihood of success of the mission and the feasibility of the design, these were both better suited to the helicopter. So, with that, it was concluded that the helicopter is best suited for this job.

2.2.2 Vehicle Design

Having come to a conclusion of what vehicle we were going to use, the second week would be the starting point of the design of the recovery helicopter. The first half of the week was designated to the baseline configuration. This would involve initial work on what the vehicle would look like, so the configurations of the rotors, empennage, fuselage, skids etc. Another key decision taken was that of having an unmanned aerial vehicle (UAV). During the second half of the week, the individual sub teams would begin work on their individual parts, this would include making a start on the initial sizing and conceptual design of the vehicle. Work on the initial system layout, the recovery mechanism, initial fuselage sizing, initial rotor and thrust sizing, powerplant integration and mission profile were conducted during the second half of the week.

The third week was when work on the preliminary design began. First of all, the maximum take-off weight final iteration was conducted. This was followed by the fuselage drag, rotor aerodynamic and structural design, fuselage design and systems integration.

The fourth week would essentially be a continuation of the work from week 3. The final fuselage design was chosen, the blade design was further worked on, static and longitudinal stability was done, and the cameras were installed around the vehicle. Initial work on the ground control station and its configuration was also done this week.

The final week was spent finishing off the design. We began performance analysis and the simulation team started flying the UAV. The full design of the synchropter was finished on Thursday night. This would allow the weekend to be spent on the final report.

2.3 Communication With The Launch Team

For our project to be successful, it was paramount that communication between ourselves and the launch team was to a high standard. As this is a wholly reusable launch system, we have to recover all stages of the rocket, unless they specify they are landing an individual stage themselves. The vehicle selection, sizing and design of the vehicle relied entirely on the parameters we were given by

the launch team. This proved to be challenging and, at times, disruptive to our project.

The first communication problem arose during the vehicle selection process. Each of the 3 concepts had a specific range they could operate between for a specific time. Our best option, the helicopter, had a maximum operational range of approximately 50 km for the time of recovery we had, therefore, we needed confirmation that they could guarantee the payload to be within this range, otherwise we would not be able to operate the helicopter. We were told by one of the members of the aerothermodynamics team working on this case that they can indeed guarantee a maximum range of 50 km for the payload, however, this was overturned by one of the coordinators, telling us that we can simply use this value “for now” but it may change. This, for us, is not a good enough answer as we must make a decision so we can start work on the design, and we cannot have an uncertainty as this could put the whole mission in jeopardy if we design a helicopter with a max range of 50 km with the actual range being greater than this. After a few days of back and forth conversations, the coordinator of the launch team told us that what he meant by saying “for now” is that the 50 km range is not fixed and is bound to change, however it won’t be greater than 50 km. We clearly stated to him that our maximum range is 50 km, and at no point did we mention that we needed an exact value, yet this was his argument. This cost us a couple of days that could have been spent on the design of the vehicle.

Another communication difficulty we had was for the fairings. It was not made clear to us whether we would be recovering the fairings individually or with the second stage. We found out that the fairings will be recovered with the second stage through their page, by chance, when we are not supposed to be there, nothing was communicated to us. Had we needed to recover the fairings separately, we would have needed an extra helicopter and boat in a best-case scenario, or 2 helicopters and 1 boat in a worst-case scenario. The reason for this is that the boat would not be able to travel the required distance between the separate captures. This would at least double the cost of the mission, which is why it was paramount that we got this information.

Furthermore, it was determined that if the inclination of the rocket is greater than 30 degrees, we would need 2 boats and 2 helicopters. This information was not passed on to us. Again, the same problems arise with this case as mentioned in the previous paragraph.

For us to reasonably determine the maximum take-off weight of our vehicle, we need to know how much the maximum weight we will be recovering is, as the weight of the helicopter depends on this. The launch team could not provide us with a final value to work with, they kept changing the weight of their first stage (the heaviest stage) which was something we could not afford as we needed to calculate to maximum take-off weight immediately. We kept getting different values as to what the mass would be, however this was not optimal for us. This showed a lack of effective decision making from the launch team as the constant change in weight caused us to repeat the same calculations multiple times.

Finally, towards the end of the project, we were getting all sorts of different values for the lengths of the stages we were recovering. We went on their slack and found a diagram that clearly labelled all the different lengths of the rocket. However, these were different to the values they gave us. So instead of them updating us on what the latest lengths were, we had to get in touch with them. Although this is more on the minor side, had we not asked for the most recent lengths, we may have ended up recovering something that was not the size we expected it to be.

Where we believed there to be an issue that did not suit our design, we would not simply accept this. Meetings were organised throughout the project with the launch team to sort these issues and come up with a solution that suits both teams.

2.4 Key Engineering Decisions And Conflict

As project managers, it is our job to make key decisions on the design. Given the magnitude of the project, 5 weeks to complete it is not nearly enough, therefore, on a few different occasions, we had to step in and analyse the situation to determine if this was a viable time period to complete the required work. If this were not possible, we would conduct a risk assessment and make a compromise on the design to allow for the task to be completed in the given time frame. Throughout the project, there have been design parameters that would benefit one sub team but be troublesome for another, and vice versa. This again was our job to make an appropriate decision that would benefit the design most and maximise the possibility of completion of the design in our 5-week time frame. Problems with individuals throughout the course of the project were minimal, however, there was 1 case where we had issues with a member of one of the sub teams.

The first major decision we had to make was whether or not we were going to use a back rotor. This idea was highly encouraged by the aerodynamics team as they believed it would significantly better the performance of the helicopter by having extra manoeuvrability and better stability, however, the flights mechanics and control team told us they would not be able to model this in the 5 weeks we had as this required some complex and very long analysis. Myself and the other coordinators decided that although having the back rotor would result in a much better design, if it was not complete in time, we would not have a design that flies, so we decided to design without a tail rotor, a compromise we believed was more beneficial for the success of the project. It was also reported to us that although the extra manoeuvrability would help ease the recovery, it was not necessary for a successful capture.

The second key decision we had to take was the weight of the payload we used to size our vehicle. The launch team reported to us that the weight of their heaviest stage would be between 1000 and 1400 kg. This was a significantly large uncertainty for us to work with, as the weight of the helicopter varies significantly with a 400 kg uncertainty. We were told to wait another few days to get the exact value of this weight, however, it was decided that this would set us back a long way and would jeopardise the completion of the project, so we decided to go ahead and design for the most constraining case. Although designing for the most constraining case would result in the heaviest design of the helicopter, and hence, a more expensive helicopter, had we opted for anything less, this would have left the possibility of our helicopter not being able to deal with the worst case weight we were given. This would cause the mission to fail as recovery would not be possible.

At the beginning of the fourth week, we were contacted by a member of the aerodynamics team telling us that he had made a mistake in the sizing of the blade at the beginning of the project. There were 2 ways we could go about this, either we redo all the calculations up to the point or we increase the disc loading from 6 to 9.4 [6]. Such an increase of disc loading would result in the efficiency of the helicopter to significantly decrease. In fact, a disc loading of 9.4 corresponds to an extremely inefficient helicopter. Although this takes a big toll on the quality of the design, the recovery can still be conducted, however, redoing all the calculations with less than 2 weeks before the deadline would simply not allow us to finish the project. Therefore, we decided to go with increasing the disc loading.

When it came to the fuselage shape selection, there was 3 final shapes presented to us [7, 8, 9]. Some of these had better aerodynamic performance than others whilst having a worst drag or pitching moment etc. The position of the inlet also had a major impact in the final selection [10]. We believe this to be more of an aerodynamic decision rather than an engineering one, so we gave the aerodynamics team the initiative of choosing which compromises are best.

Throughout the project, communication between our team was at a high standard, however, this was not the case for one of the team members. This individual failed to communicate with us throughout the duration of the project, meaning we did not know at any point what his contribution to the team was. Towards the tail end of the project, one of the flight mechanics guys was working on a code for the performance part, as he was told no one else would be working on this. A few days later, when this code was completed, we received a code that calculated the same thing from this other individual. This was simply due to the lack of communication from this person, and given the fact we were having some blade issues at the same time this happened, it would have been a lot more optimal for him to be working in this than on something that has already been done. We had a meeting with this individual and simply told him that this was not acceptable, and we demanded a greater input from him into the communication of his duties.

3 Regulations

As our helicopter is an unmanned aerial vehicle without any passengers, the corresponding regulations for such vehicle are those for a category B rotorcraft from FAR 29 [11]. These regulations are for rotorcrafts with a weight less than 20,000 lbs and less than 9 passengers. Other regulations we had to comply with were the FAR 133 regulations. These were specific to external cargo carriers [12]. To facilitate access to these regulations for the project, I created a short summary of the key constraints needed to design the helicopter. As the regulations are hundreds of pages long, I decided not to include everything, as this would be 10s of pages of summary. I told the team that if they had any questions regarding the regulations or if there was anything they needed that was not in the summary list I created to contact me and I would let them know whether or not these regulations were needed.

FAR 29 is split into 6 main subparts each split into different sections: flight (general, performance, flight characteristics, ground and water handling), strength requirements (general, flight loads, control surface and system loads, ground loads, water loads, main component requirements, emergency landing), design and construction (general, rotors, control systems, landing gear, floats and hulls, personnel and cargo accommodations, fire protection, external loads), powerplant (general, rotor drive system, fuel system, oil system, cooling, induction systems, exhaust system, controls and accessories, fire protection), equipment (general, instrument installation, electric systems, lights, safety equipment) and operating limitations and information (operating limitations, markings and placards, rotorcraft flight manual).

3.1 Neglected FAR 29 Regulations

As we were operating an unmanned category B rotorcraft without any passengers, many regulations were not needed to be complied with. I will list all regulations that were excluded and give the reasoning behind it.

Flight: All take-off and landing parameters corresponding to category A rotorcrafts were neglected as we are operating a category B rotorcraft. One engine inoperative (OEI) climb was also neglected as this requires the rotorcraft to have multiple engines, ours only has one. For this reason, all regulations including multiple engines and category A rotorcrafts were neglected. As we do not have a back rotor, regulations regarding this were ignored.

Strength requirements: For this subpart, limit pilot forces (forces acting on the pilot from controls such as the foot pedals and joystick) were neglected as the vehicle is unmanned. All regulations regarding tires, shock absorbers, wheels and skis were ignored as our vehicle consisted of skids and not wheels. As our vehicle is not designed to operate on water, there was no need to comply with water loads.

Design and constructions: Although in real life the vehicle would need to go through a lot of tests to make sure its operable, this is something we were not going to be doing as we are not actually building the helicopter, so regulations regarding the physical testing of various aspects of the vehicle were not taken into account, however, in real life application, these are compulsory to developing an aerospace vehicle. For the same reasons as previously mentioned, all landing gear regulations not related to skids were neglected. Floats and hulls were also neglected since the vehicle does not operate on water, as previously mentioned. All personnel and cargo accommodation regulations such as pilot and cargo compartments and emergency evacuation were neglected as we do not have any personnel or interior cargo. Manual fire extinguisher regulations were ignored for obvious reasons.

Powerplant: Fuel tank, oil tank and cooling tests regulations were not taken into account for the same reasons mentioned in the above paragraph. As we are using a turbine engine, all other types of engines were ignored. As mentioned before, all regulations on physical tests for this part too were neglected.

Equipment: As there is nobody on board the helicopter during operation, all regulations for human safety equipment such as seatbelts, oxygen masks etc were neglected. Cockpit voice recorders were also left out.

Operating limitations and information: Since we were not actually building the helicopter, all regulations on markings and placards were ignored as the design did not require any of this. These would not be ignored when building the vehicle as they are paramount to be able to distinguish between different characteristics.

3.2 FAR 29 Key Constraints

In the following paragraphs, I will be summarising the key constraints of each subpart that the team had to comply with. (these are not all the regulations we must comply with, just the key ones that affect the design the most).

Flight: Performance must be based on relative humidity of 80 percent at and below standard temperature and at 34 percent at and above standard temperature plus 50 degrees F. Between these 2, relative humidity varies linearly. Satisfactory roll and pitch control must be demonstrated with power off at 1.1 VNE (never exceed velocity). VNE must be greater than 40 knots and less than 0.9 times the maximum forward speed. Static longitudinal stability must be shown at speeds from $V_y - 10\text{kt}$ to $V_y + 10\text{kt}$ (V_y is the speed that allows for the best rate of climb) in climb and at speeds

from $0.8 \text{ VNE} - 10\text{kt}$ to $0.8 \text{ VNE} + 10\text{kt}$ or if V_h is less than 0.8 VNE , from $V_h - 10\text{kt}$ to $V_h + 10\text{kt}$ (V_h is max speed in level flight at max continuous power) in cruise. Static longitudinal stability must also be shown at speeds from $\text{VNE} - 20 \text{ kt}$ to VNE . All these values must be shown for the most constraining cases. For static directional stability, sideslip angles must increase with steadily increasing directional control deflection for sideslip angles up to ± 25 degrees from trim at a speed of 15 knots less than the speed for min rate of descent varying linearly to ± 10 degrees from trim at VNE .

Strength requirements: The rotorcraft must be designed for a limit load factor ranging from a positive limit of 3.5 and a negative limit of -1 and must be designed to withstand the loads resulting from vertical and horizontal gusts of 30 feet per second. The limit engine torque may not be less than the mean torque for maximum power multiplied by 1.25. For specified landing conditions, a design maximum weight must be used that is not less than the maximum weight. The rotor lift may not exceed $2/3$ of the design max weight.

Design and construction: Each critical casting must have a casting factor of no less than 1.25. Fitting factors of at least 1.15 must be applied to each part of the fitting. The rotorcraft must be designed to ensure capability of safe landing after impact with a 1 kg bird when the velocity is equal to the lesser of V_h or VNE at altitudes up to 8000 ft. Although this is not an issue for our mission, if we wish to sell the helicopter after the mission to be reconverted for other use, it must comply with this regulation. The rotorcraft external load attaching means for a rotorcraft-load combination to be used for non-human external cargo application must withstand a limit static load of 2.5 multiplied by the maximum external load.

Powerplant: Regarding the crash resistance of the fuel tanks, they must be able to withstand 1.5g in the upward direction, 4g in the forward direction, 2g sideward and 4g downward. Although this is primarily for preventing a fire for passenger safety, they must be complied with in case of a crash landing which could ignite a fire on the boat. Each oil tank used must have an expansion space not less than 10 percent of the tank capacity.

3.3 FAR 133

This part of the regulations corresponds to external load operations rotorcrafts and is split into 3 main subparts: certification rules, operating rules and related requirements and airworthiness requirements. None of these regulations turned out to be constraints for the design of the helicopter as they were mainly about the prohibition of the carriage of drugs, the employment of personnel etc. These would have to be complied with when the vehicle is built, but since we are not building the vehicle, we decided to dedicate our time to more important things given the tight 5 week schedule of the project.

4 UAV Ground Control Station

Unmanned aerial vehicles are becoming more common by the year. Whether its drones aiding with construction work, package deliveries or even just pleasure flying on a sunny day at the park, they have greatly evolved and are used worldwide today. The majority of these require a simple hand operated controller with minimal technology. This is not the case with our design. Our vehicle is designed to recover a 1.4 tonne payload in a multi million dollar mission, which would make it the second largest Unmanned aerial vehicle in today's world in terms of payload capacity behind the

Feihong-98 which can recover payloads of up to 1.5 tonnes [13]. This, for obvious reasons, cannot simply be controlled using a hand controller, but will require a complex ground control station instead. 2 options were considered for this, a stationary and a portable ground control station. I will explain the setup of both these configurations, and which one suits our mission most.

4.1 Primary Controls

A conventional helicopter consists of 4 primary controls, a cyclic pitch control, collective pitch control, anti-torque pedals and a throttle [14]. Our ground control station will not have anti-torque pedals as these are used to control the back rotor, which we do not have.

The cyclic pitch control is split into 2 different configurations, longitudinal and lateral control[14, 15].

Longitudinal: This varies the main rotor blade pitch with fore and aft movement. Its primary effect is to tilt the main rotor disk forwards and back via the swashplate. Its use in forward flight is to adjust the forward speed and to control the rolled turn. In hover, its used to eliminate forwards and backwards movement[14, 15].

Lateral: this has the exact same use as the longitudinal control but sideways. Its primary effect is to tilt the main rotor disc left and right, its forward flight use is to create lateral movement and in hover it is used to eliminate sideways movement[14, 15].

The collective stick is operated in the same was as a conventional joystick. Forward and backward movement translates to negative and positive pitch respectively. Left and right movement of the stick translates translates to sideways movement[14, 15].

The collective pitch control is used to change the pitch angle of all main rotor blades collectively causing the helicopter to climb or descend. Its use in forward flight is to adjust the power through the rotor blade pitch setting and in hover its used to maintain the desired altitude. This is simply operated by moving the control lever up and down[14, 15].

The throttle is used to control the power of the engine by regulating its rpm. This ensures the blades generate enough lift for the helicopter to stay airborne throughout the flight. The throttle control is usually mounted on the collective control stick. The rpm increases with anticlockwise rotation and decreases with clockwise rotation[14, 15].

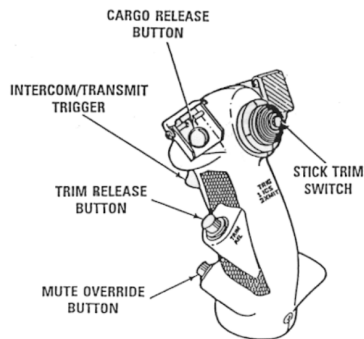


Figure 1: cyclic stick to be used for the GCS [1]

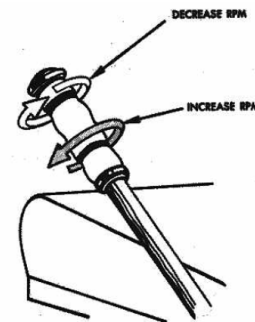


Figure 2: Collective stick to be used for the GCS [2]

4.2 Key Instruments To Be Used

The same flight and engine instruments will be used as those in a conventional modern-day helicopter. The flight instruments consist of an artificial horizon, air speed indicator, altimeter, vertical speed indicator, turn and bank indicator and a compass [3].

The engine instruments will consist of 3 main instruments: A dual tachometer to indicate the number of revolutions per minute of both the engine and rotor blades, a manifold pressure gauge to indicate the amount of work the engine is doing and a torque meter to measure the torque on the engine and rotor [4].

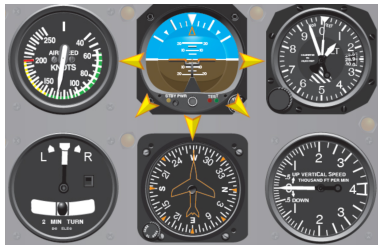


Figure 3: Flight Instruments for the GCS [3]

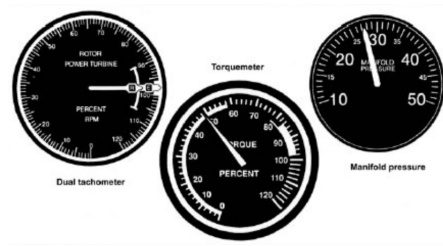


Figure 4: Engine instruments for the GCS [4]

As the sole purpose of this mission is to recover a payload in the middle of the ocean from a boat, the ground control station will not need to be designed to have a radar, the only GPS instrument that will be required will be one that shows the relative position of the UAV from the boat.

The pilot will have a screen that allows him to have access to all camera views of the vehicle. There will be 6 cameras mounted around the fuselage. 5 of those are ruggedvid HD29mm with a 95 degree field of view (FOV) and 1 is an AVT CM62 gimbal mounted camera at the nose of the helicopter with a 180 x 180 degree longitudinal – lateral FOV and 50x optical zoom. 1 meggit CMZ camera is mounted on the belly of the vehicle with a 48 degree FOV. This will be particularly useful for landing and the recovery process [5].

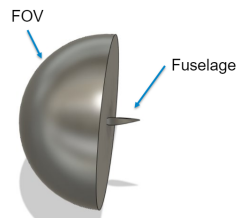


Figure 5: FOV of the nose mounted camera [5]

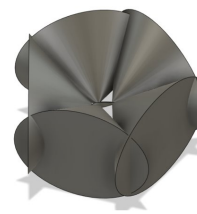


Figure 6: FOV of the 5 ruggedvid cameras [5]



Figure 7: FOV of the belly mounted camera [5]

4.3 Ground Control Station Setup

4.3.1 Stationary Control Station

The stationary control station will be set up such that the pilot is seated with a 180 degree screen around him to mimic an actual helicopter cockpit. The view displayed by this screen will be dictated by the gimbal nose camera. The cyclic stick will rise from beneath the front of the pilots seat, and the collective stick will be to the left of the pilots seat, with the throttle control at the tip of the stick. To the right of the pilot will be a 50 x 40 cm screen which will be used to access the views provided by the other 6 cameras. The pilot can choose whichever view he likes at any point during the flight. Having such flexibility will be particularly useful during the recovery stage, as this gives the pilot a lot more angles to work with than a manned recovery vehicle. To the left of the pilot will be a small GPS device that displays the relative position of the UAV to the boat and directly opposite the pilot will be the flight and engine instruments.

There are many positive aspects to this particular setup: A virtual reality setup as such will allow for a near identical experience had you been operating the helicopter on board. The control station can be designed to have identical size instruments and controls to that of an actual cockpit due to the freedom of space making it more realistic case. This configuration also allows the pilot to have 2 different views simultaneously from the extra screen. This will be handy during key manoeuvres such as recovery and landing.

As great as this setup might seem, there is also some limitations to it. As the operation will happen in the middle of the ocean and the station is on land, the pilot will never have a visual on the UAV, which can be particularly redundant in case of a system failure. A failure of one of the cameras, or more specifically, the front or belly camera would essentially end the mission.

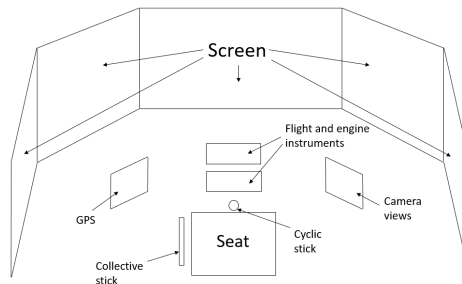


Figure 8: Stationary ground control station setup

4.3.2 Portable Control Station

The portable control station will be set up in a 1.5 x 0.4 m case. On the vertical face of the case will be a 0.6 x 0.4 m screen at the centre that displays the views from all 7 vehicle cameras. The pilot will only have access to one of these views at a time. To the right of the screen will be the GPS showing the position of the helicopter to the boat and to the left will be the flight and engine instruments. All these will operate in the same fashion as for the stationary station. On the horizontal face of the case will simply be the collective control stick to the left and the cyclic control stick to the right.

The benefit of having a portable control station is that the vehicle can be operated from absolutely anywhere in the world. This gives us the ability to take the case on board the boat and operate

from there which will give the pilot a visual reference with the UAV during take-off, landing, and potentially even the recovery if it happens close enough to the boat.

The main limitation of this configuration is that the pilot only has access to 1 camera view at a time, and unlike the main screen on the stationary station which allows you to have the same view as a real cockpit, the screen on the portable station is relatively small. Having access to only 1 camera at a time is not a big issue for normal operating flight, however, when attempting to recover a 1.4 tonne rocket stage travelling at high speeds in more than 1 direction, this could make the recovery process extremely difficult, especially with the small size of the screen. Portable stations are usually designed for small UAV's and given the magnitude of the mission and size of the vehicle, it could prove challenging developing a system to accommodate for this.

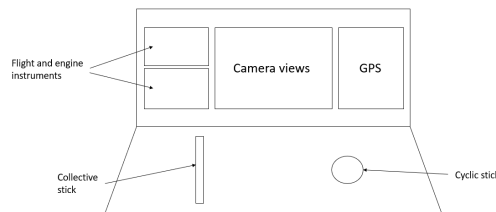


Figure 9: Portable ground control station setup

4.4 Stationary Vs Portable Station

After taking into consideration all aspects of both configurations, it was decided that the stationary ground control station is best suited for this mission. The benefits of the stationary system significantly outweigh those of the portable station. Having a visual reference can make the mission easier, however, the recovery is very much doable without this. The access to a lifelike screen system and 2 camera views simultaneously will increase the likelihood of success of the mission greatly and each camera will have multiple backup systems to ensure it does not fail at any point during the mission profile.

5 Conclusion

Overall, i think the decision to give all 3 project coordinators equal input into the management of the team proved optimal. This allowed for second and third party opinions on key decisions which eliminated any bias towards the design. I learned that to make such a high profile mission successful, a lot more than 5 weeks is required. This time constraint challenged myself and the other coordinators to take on key decisions and make adequate design compromises to ensure the completion of the project in the given time frame. I believe as coordinators, we effectively kept the group on track throughout the duration of the project and we dealt with issues in an adequate and professional way. In the future, the blade design will need to be done more extensively as this proved to be the weak spot of the project. Regrading regulations, although there is hundreds of different regulations a helicopter must comply with, many were simply not needed for our project as these require the vehicle to actually be built, which is not the case for our project. Designing an appropriate ground control station is paramount to ensure a successful mission, therefore a design similar to that of a cockpit was optimal for our mission.

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A Appendices

A.1 Vehicle Selection Decision Matrices

Weighting Matrix

Parameter	Time for capture	Speed/Range	Cost	Likelihood of success	Pilot workload	Feasibility of design	Maneuverability and response	Normalized mean
Time for capture	1.00	3.00	0.20	0.14	8.00	0.14	1.00	0.44
Speed/Range	0.33	1.00	0.33	0.20	6.00	0.20	0.50	0.28
Cost	5.00	3.00	1.00	1.00	10.00	1.00	3.00	0.79
Likelihood of success	7.00	5.00	1.00	1.00	10.00	2.00	4.00	0.98
Pilot workload	0.13	0.17	0.10	0.10	1.00	0.10	0.13	0.06
Feasibility of design	7.00	5.00	1.00	0.50	10.00	1.00	6.00	1.00
Maneuverability and response	1.00	2.00	0.33	0.25	8.00	0.17	1.00	0.42

<1 is low importance, >1 high importance

Figure 10: Weight matrix of relative importance between the different design drivers

Risk Factors for Each Concept

Parameter	Aircraft	VTOL	Helicopter
Time for capture	10.00	2.00	1.00
Speed/Range	1.00	5.00	10.00
Likelihood of success	6.00	4.00	2.00
Feasibility of design	3.00	6.00	4.00
Average	5.00	4.25	4.25

Note: 1 is low risk, 10 high risk

Figure 11: Risk factors for each concept

A.2 CAD



Figure 12: CAD model of the unmanned synchrocopter